

Quality Inspection Report QIR-2026-0031

Dimensional / Lot Acceptance Inspection | LINER HANGER ASSY, FINAL ASSEMBLY INSPECTION

Report No. / Date	QIR-2026-0031 / 2026-02-11
Part Number	ACT-LH-7625
Part Description	LINER HANGER ASSY, FINAL ASSEMBLY INSPECTION
Work Order / PO	WO-26-0087 / SO-31022 (Permian operator)
Lot / Heat-Batch	- / Body: K73209 (13Cr)
Lot Qty / Sample Qty	2 / 2 (per ACT-QP-08)
Serial Numbers	LH76-26-0007 / LH76-26-0008
Material	Per BOM-LH-7625 Rev A
Inspected By	D. Whitfield
Reviewed / Approved By	S. Nguyen
Applicable Specs	API 11D1 / API 19V (as applicable), NACE MR0175/ISO 15156, ASTM E709, API 5B

Dimensional & Attribute Results

Actuals are the worst-case (closest-to-limit) reading across the sample unless noted. Instruments are calibrated per ACT-CAL-01; calibration records on file.

Char.	Feature / Characteristic	Nominal	Tol +	Tol -	Actual (worst)	Instrument	A/R
D1	Max assembled OD	6.50	+0	--0.01	6.51	Ring gauge R-65	ACC
D2	Min through-ID (drift)	3.920	+0.010	-0	3.927	Drift bar DB-392	ACC
D3	PBR bore ID	4.000	+0.002	-0	4.001	Air gauge A-400	ACC
D4	PBR surface finish (Ra uin)	16.0	+0	--8.0	24.0	Profilometer P-52	ACC
D5	Slip standoff (retracted)	0.125	+0.020	--0.020	0.145	Feeler/depth DM-3	ACC
D6	Shear screw protrusion	0.000	+0.005	-0	0.003	Depth mic DM-3	ACC
D7	Setting cylinder function test (psi to first movement)	400.0	+100.0	--100.0	500.0	Test bench TB-2	ACC

Result: ACCEPT

Measuring & Test Equipment Used

Instrument	Cal. Record	Cal. Due	Status
Ring gauge R-65	CAL-5057	2026-06-01	In calibration at time of use
Drift bar DB-392	CAL-3790	2026-03-12	In calibration at time of use
Air gauge A-400	CAL-3339	2025-05-16	In calibration at time of use
Profilometer P-52	CAL-7097	2025-01-28	In calibration at time of use
Feeler/depth DM-3	CAL-5628	2025-03-03	In calibration at time of use
Depth mic DM-3	CAL-0531	2026-10-20	In calibration at time of use
Test bench TB-2	CAL-9519	2026-04-24	In calibration at time of use

Sampling & Methods

Sampling per ACT-QP-08 (ANSI/ASQ Z1.4 basis). Dimensional measurements at 68-72 degF after minimum 4 hr thermal soak. Surface finish per ASME B46.1. Hardness per ASTM E18 (HRC/HRB) or ASTM E140 conversion where noted. Threads gauged per API 5B with certified working gauges. Magnetic particle inspection, where applied, per ASTM E709 using wet fluorescent method, 100-125 V/in. field.

Inspector Notes

Both assemblies passed final inspection and function test. Hydrostatic body test per HTR-2024-0389 protocol at 8,000 psi. Serial LH76-26-0008 shipped with cert package CP-26-0102 including MTR heat K73209.

Sign-off

Role	Name	Signature	Date
QC Inspector (Level II)	D. Whitfield	/signed electronically/	2026-02-11
QA Manager	S. Nguyen	/signed electronically/	2026-02-11